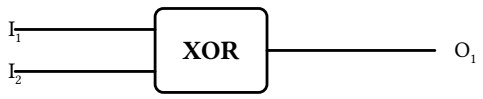
Logic Gate Exercises

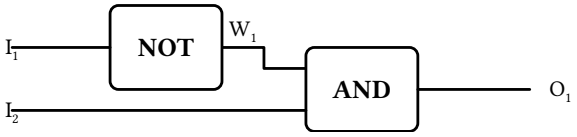
I = input, O = output, W = internal wire. Fill in the shaded cells for every combination of inputs.

Exercise 1



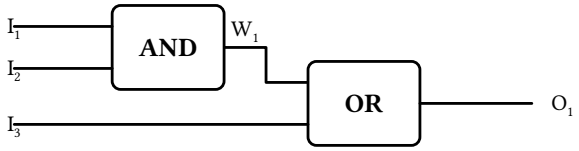
I_1	I_2	O_1
0	0	0
0	1	1
1	0	1
1	1	0

Exercise 2



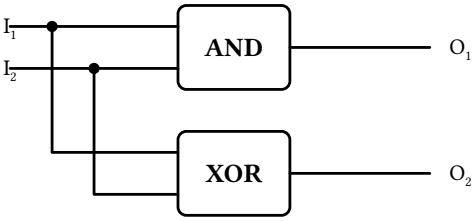
I_1	I_2	O_1
0	0	0
0	1	1
1	0	0
1	1	0

Exercise 3



I_1	I_2	I_3	O_1
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Exercise 4



I_1	I_2	O_1	O_2
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Solve Operations:

- $\neg(100100)_2$
- $(101011)_2 \wedge (100100)_2$
- $(101011)_2 \vee (100100)_2$
- $(101011)_2 \oplus (100100)_2$

Problem: When is $x \vee y = c$ solvable?

You are given an arbitrary integer $x \in \mathbb{N}$, and an integer $c \in \mathbb{N}$. You can choose any $y \in \mathbb{N}$. Let:

- $a = (a_1, a_2, a_3, ..., a_{S-1}, a_S)$ be the list of bits of x
- $b = (b_1, b_2, b_3, ..., b_{S-1}, b_S)$ be the list of bits of c
- $\vee$ denote bit-wise OR on any two numbers.
- All numbers have the same bit-length, and S be the bit-length.
- a_i and b_i denote the bit at the i-th position ($1 \leq i \leq S$)
- $a_i, b_i \in \{0, 1\}$

What conditions must be satisfied at a_i and b_i such that: $x \vee y = c$?